

Which Uncertainty Is Worth Reducing? Budgeted Measurement for Wellbeing Prioritisation

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Abstract

Workforce wellbeing surveys can guide which organisations receive closer review, but collecting more responses need not resolve uncertainty about that decision. We study a fixed budget for additional measurement before allocating a fixed number of review slots. An ordinary channel may share a persistent organisation-specific offset; a more expensive validation channel measures a reference target only under an explicit validity assumption. We specialise existing Gaussian knowledge gradient to this decision and distinguish three phenomena: irreducible ranking uncertainty from unit-specific offsets, harmless shared shifts under within-group quotas, and finite-budget losses from costly or invalid validation. Controlled Gaussian experiments show both gains and reversals. A temporally separated benchmark uses 206,098 employed-adult BRFSS records from 2023 for calibration and 214,151 from 2024 for hidden-record replay across 48 areas. Under a controlled recruitment tilt and validation cost five, two-channel knowledge gradient increases severity regret from 0.1615 to 0.1857 at budget 144, despite slightly reducing estimation error. NHS aggregate data provide a separate conditional illustration, not an intervention evaluation. The contribution is an auditable decision benchmark and a characterisation of when additional measurement can fail; neither a new acquisition algorithm nor improved employee health is claimed.

1 Introduction

A workforce team with limited capacity may need to decide which organisations warrant a closer wellbeing review. The organisations with the lowest reported scores are not necessarily those with the greatest underlying need. Sampling noise, differential participation and changes in the measurement process can alter the list. A natural response is to collect more information before deciding. Yet an additional survey through the same recruitment channel can repeat the same distortion, while a costly independent assessment can consume the budget before resolving enough comparisons. The practical question is therefore not simply where uncertainty is largest, but which additional measurements can change a consequential decision.

The NHS provides a concrete motivation, with important limits. The annual NHS Staff Survey and quarterly staff-listening infrastructure inform workforce discussions. National Quarterly Pulse Survey guidance preserves an opportunity for all staff to respond in relevant rounds [NHS England, 2023b]. Targeted acquisition must therefore mean *additional* follow-up on top of universal opportunity. It must not remove routine invitations from organisations judged low priority. Moreover, quarterly engagement items are not interchangeable with the annual burnout composite. Our proposed users are staff-experience leads and workforce analysts making a cross-organisation review decision; no NHS stakeholder has yet validated this workflow.

This study examines the measurement stage, not allocation of treatment or financial returns from employee wellbeing programmes. A review slot means capacity to examine a unit’s situation and discuss next steps. Lower target scores indicate greater need. We minimise the difference between the average target score of the selected units and that of an oracle selection with the same capacities. This severity loss distinguishes an almost-tied substitution from overlooking a much worse unit. It also allows clear analysis of when uncertainty about a common level shift is irrelevant to a fixed within-group decision.

Our method uses established Gaussian knowledge gradient (KG) and multi-information-source optimisation (MISO), not a newly invented acquisition principle [Frazier et al., 2008, 2009, Poloczek et al., 2017]. The study contributes an NHS-informed decision specification, an explicit account of which uncertainty affects that specification, and a reproducible benchmark that can reject an appealing measurement policy. The benchmark separates model-generated latent truth from real recorded outcomes with controlled sampling. In the latter, calibration uses the previous year and policies cannot access the hidden current-year targets. All policies share the same inference model except designated bias-blind baselines.

Three findings challenge a blanket recommendation to validate more. First, persistent offsets matter through relevant *differences*: independent unit offsets can leave selection regret even after unlimited ordinary measurement, whereas a shared within-group offset cancels from this loss. Second, myopic gain per cost does not solve the multistep budget problem. Third, on real-record replay, lower mean squared estimation error can accompany worse review selections. These findings support a limited social-impact claim: the framework can expose unjustified measurement expenditure before a field deployment. It does not establish that its recommendations improve NHS decisions or employee health.

2 Related Work

Gaussian KG values a measurement by the expected improvement in a subsequent decision [Frazier et al., 2008, 2009]. MISO explicitly models discrepancies between cheaper information sources and an expensive target source [Poloczek et al., 2017]. Our independent two-dimensional unit model is a small, transparent instance of that programme. The cost-normalised rule remains myopic; nonconcavity of information value can make sequential cost ratios misleading [Frazier and Powell, 2010].

Optimal-subset allocation, fixed-confidence subset identification and combinatorial pure exploration have different objectives and guarantees [Chen et al., 2008, Kalyanakrishnan et al., 2012, Chen et al., 2014]. We study expected severity regret under a fixed budget. Our boundary-directed baseline is a heuristic, not an implementation carrying a LUCB sample-complexity guarantee. Contaminated observations can prevent identification even with unlimited samples [Alschuler et al., 2019]. Multi-fidelity best-arm identification and recent selective proxy auditing also address cheaper biased sources [Poiani et al., 2024, Ao et al., 2026]. We do not claim the existence of a second channel, the Gaussian conditioning formula, or persistent-bias nonidentification as new.

AI for social impact includes resource-constrained data acquisition. Betti et al. [2026] study spatial mapping on a budget, and Goebel et al. [2026] study budgeted active learning using expert advice and episodic priors. Our distinction is between noise, persistent channel error and decision-invariant shared shifts in a review-selection task. Health-service profiling already cautions against overinterpreting ranks [Goldstein and Spiegelhalter, 1996, Lin et al., 2006]. Here the intervention being evaluated computationally is a *measurement policy*; it is not a wellbeing intervention or a causal estimate of organisational benefit.

3 Decision and Measurement Model

There are M units partitioned into comparison groups g , with prescribed review capacities k_g . Let $\mathcal{S} = \{S : |S \cap g| = k_g\}$ and $K = \sum_g k_g$. For lower-is-worse target θ_i , define

$$L(S, \theta) = K^{-1} \left(\sum_{i \in S} \theta_i - \min_{A \in \mathcal{S}} \sum_{i \in A} \theta_i \right). \quad (1)$$

Each selected unit has equal weight. Capacities are a decision assumption, not a proven fairness criterion. Conditional on observed data, the Bayes action selects the k_g smallest posterior target means in each group. Linearity of expectation establishes this decision even for dependent targets.

An ordinary measurement packet observes $Y_{iR} = \theta_i + b_i + \epsilon_{iR}$; an ideal validation packet observes $Y_{iA} = \theta_i + \epsilon_{iA}$. A persistent b_i is shared across repeat packets from unit i . Packets have known costs c_R, c_A and fresh conditional noise variance r_{id} . At each step an affordable unit-channel action is selected, its observation updates the posterior, and its cost is deducted. The final set is chosen when no action of that policy is affordable. Initial observations are common to every policy and excluded from the additional budget.

For acquisition, assume independent unit states and fixed hyperparameters:

$$(\theta_i, b_i) \mid D \sim N \left(\begin{pmatrix} m_i \\ a_i \end{pmatrix}, P_i = \begin{pmatrix} v_i & u_i \\ u_i & q_i \end{pmatrix} \right). \quad (2)$$

The loading is $h_R = (1, 1)^\top$ or $h_A = (1, 0)^\top$. Gaussian conditioning gives

$$P_i^+ = P_i - \frac{P_i h_d h_d^\top P_i}{h_d^\top P_i h_d + r_{id}}.$$

The random change in target posterior mean has standard deviation

$$s_{id} = \frac{|(P_i h_d)_1|}{\sqrt{h_d^\top P_i h_d + r_{id}}}. \quad (3)$$

For a independent respondent draws of variance σ^2 , a packet mean has noise variance σ^2/a . Persistent channel offsets do not shrink by this factor. Validation unbiasedness is an assumption, not a consequence of using different respondents.

Exact one-step value. For $0 < k_g < |g|$, let t_i be the k_g th smallest *other* posterior mean and $d_i = |m_i - t_i|$. If the action changes only coordinate i , its expected reduction in Bayes severity risk is

$$\text{KG}_{id} = \frac{s_{id} \varphi(d_i/s_{id}) - d_i \Phi(-d_i/s_{id})}{K}, \quad (4)$$

with zero value at $s_{id} = 0$ or a trivial group quota. This is the expected positive part of a normal boundary crossing. For a currently selected unit the alternative cutoff is the group's $(k_g + 1)$ st mean, not its k_g th mean. The distinction matters in implementation. Our two-channel KG policy maximises KG_{id}/c_d over affordable actions. The $1/K$ factor is common and can be omitted for this choice. If an observation changes multiple correlated units, the one-coordinate expression is invalid.

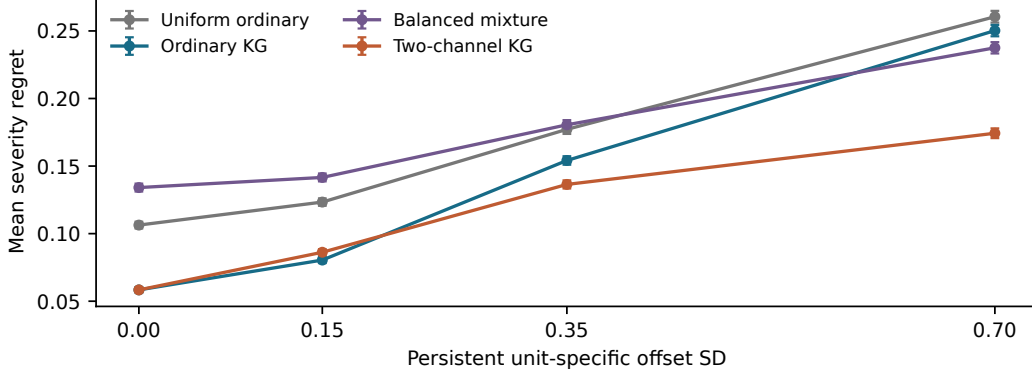


Figure 1: Controlled Gaussian worlds, budget 120 and validation cost 5. Lower severity regret is better. Error bars are 95% Monte Carlo intervals from 2,000 replications per world. Policies share the same bias-aware inference model. Two-channel KG has access to more expensive information; the balanced mixture has the same channel access.

What ordinary repetition cannot learn. For independent priors $\theta \sim N(\mu, \tau^2)$ and $b \sim N(0, \beta^2)$, infinitely many ordinary measurements leave target variance

$$V_\infty = \frac{\tau^2 \beta^2}{\tau^2 + \beta^2}. \quad (5)$$

For two independent units and one slot, even exact observations of both $\theta_i + b_i$ leave Bayes severity regret

$$R_\infty = \frac{\tau}{\sqrt{\pi}} \left(1 - \frac{\tau}{\sqrt{\tau^2 + \beta^2}} \right). \quad (6)$$

However, a positive level-variance floor does not establish selection harm. For arbitrary group constants z_g , every feasible set satisfies $L(S, \theta + z) = L(S, \theta)$. If the sole ordinary offset is shared within a group, exact ordinary observations identify its optimal set. A model that instead treats this shift as independent offsets can buy validation for a distinction irrelevant to the decision.

Validation and cost are separate assumptions. At the same unit and equal cost, with a nontrivial group quota, ideal validation has greater one-step value precisely when

$$\frac{v_i^2}{v_i + r_{iA}} > \frac{(v_i + u_i)^2}{v_i + q_i + 2u_i + r_{iR}}. \quad (7)$$

Unequal costs require the full KG ratio; variance reduction per cost is not interchangeable with it. If validation has its own persistent offset d with independent variance $\delta^2 > 0$, observing both channels exactly still leaves variance $(\tau^{-2} + \beta^{-2} + \delta^{-2})^{-1}$. Finally, greedy KG per cost need not optimise a fixed budget: the supplement gives a two-option Gaussian example where two cheap measurements yield expected improvement 0.3257, but one affordable expensive measurement yields 0.3804. These are reference-model statements, not guarantees for misspecified survey models. Full propositions, proofs and independent numerical checks are supplied in the supplement.

4 Experimental Design

Policies and outcomes. We compare ten policies. Uniform ordinary, maximum posterior variance, a boundary heuristic and ordinary-only KG share the same bias-aware update. Bias-blind uniform and KG set the assumed offset to zero. Two-channel KG, uniform sampling over unit-channel pairs, a cost-balanced random mixture, and validation-only sampling can use validation. The balanced mixture selects validation with probability $1/(1 + c_A)$ when $c_R = 1$, giving equal expected channel expenditure before budget-end effects. Two-channel KG, the equal-action mixture and the balanced mixture have the same channel access; validation-only is a restricted comparator. Comparison with ordinary KG also changes the available channels. Policies consume matched per-unit, per-channel streams. We report severity regret, selected-set overlap with the oracle, target MSE, nominal 95% interval coverage and validation spending. Coverage under misspecification is empirical, not guaranteed.

Controlled latent-truth experiments. Forty units form four groups of ten, with two review slots per group. Targets are iid $N(5, 0.35^2)$; unit-specific ordinary offsets have SD $\beta \in \{0, 0.15, 0.35, 0.70\}$. Each unit begins with 20 ordinary draws; each additional packet represents 20 draws with individual SD 2.5. Additional budgets are 40 and 120 ordinary-packet cost equivalents, with validation costs 2, 5 and 10. Each latent world has 2,000 independent replications, reused across its cost and budget settings. Seven additional worlds examine underestimated bias, overestimated bias, biased validation, heavy-tailed packet noise, unequal noise, post-initial target drift, and a shared group offset. The heavy-tail stress draws a scaled Student- t_3 *packet mean*; it is not a sample mean of 20 individual t_3 variables. All stress assumptions and the complete 310 policy-setting rows are reported.

Real-record replay, not field validation. We use official CDC BRFSS public files [Centers for Disease Control and Prevention, 2024, 2025]. Eligible records report employment or self-employment and a valid number of mentally unhealthy days. The score is $(30 - \text{MENTHLTH})/3$, with code 88 mapped to zero days; refusal, unknown and missing values are excluded. The common cohort comprises 47 states plus DC: 206,098 eligible records in 2023 and 214,151 in 2024. The primary target is each area’s *unweighted complete-file eligible-respondent mean*, not population prevalence or clinical need. Nine of 48 areas are selected. These are geographical areas, not NHS trusts or hospital departments, and this outcome is not the NHS burnout composite.

Only 2023 supplies the common target-prior centre, between-area SD and within-area sampling variances. The 2024 record distribution supplies a hidden evaluation frame. Every area initially reveals 20 draws; every new packet contains 20 draws *with replacement*. Sampling from the 31 outcome-frequency bins is exactly equivalent to record sampling for this single outcome. Repeated records are allowed: a packet is not 20 newly recruited people. Policies receive observations and frozen calibration, never the complete 2024 means.

Ordinary recruitment is either uniform or has probabilities proportional to $p_j \exp(\pm \kappa x_j / \sigma_{i,2023})$, with a persistent random sign per area and repetition. Values $\kappa = 0.25, 0.50$ are controlled mechanisms, not estimates of actual nonresponse bias. The corresponding 2023 plus/minus laws supply the working bias centre, bias variance and packet variance. This favourable knowledge of a hypothetical recruitment law is explicit. Validation is uniform except for a failure scenario with its own independent tilt of 0.25. A weighted sensitivity uses probabilities proportional to the supplied survey weights and targets the resulting *weighted finite-file distribution*; it is not design-based population inference. All five scenarios use 1,000 repetitions, budgets 48 and 144, and validation costs 2, 5 and 10 for the unweighted uniform and unweighted tilt-0.50 scenarios, or cost 5 otherwise.

Policy	Unif.	T.25	T.50	Wgt.	V.bias
Uniform ordinary	0.1397	0.1813	0.1848	0.1965	0.1848
Uniform, bias blind	0.1397	0.1929	0.2103	0.2200	0.2093
Maximum variance	0.1343	0.1766	0.1967	0.2020	0.1950
Boundary heuristic	0.1042	0.1782	0.1885	0.1996	0.1880
Ordinary KG	0.1124	0.1570	0.1615	0.1773	0.1612
Bias-blind KG	0.1124	0.1894	0.2008	0.2129	0.1994
Equal-action mixture	0.1697	0.1852	0.1893	0.1990	0.2081
Balanced mixture	0.1558	0.1863	0.1884	0.2008	0.1998
Validation only	0.1765	0.1866	0.1899	0.1995	0.2114
Two-channel KG	0.1124	0.1755	0.1857	0.1959	0.2226

Table 1: Mean severity regret on real-record replay, additional budget 144, validation cost 5, 1,000 repetitions per scenario. Columns: uniform ordinary; tilt 0.25; tilt 0.50; weighted finite-frame tilt 0.50; and ordinary tilt 0.50 with validation tilt 0.25. All targets are fixed-frame recorded means. Lower is better; this table is not a population or NHS effect estimate.

This gives 180 policy-setting rows. The posterior is a working Gaussian approximation to bounded, asymmetric observations and a two-sign bias mechanism.

NHS conditional illustration. The 2025 release provides five-year histories for a cohort of 190 trust reporting units (189 legal entities), with 36 hypothetical within-group review slots [NHS Staff Survey, 2026, NHS Staff Survey Coordination Centre, 2026]. A scenario analysis uses observed 2025 burnout scores, assumed $\tau = 0.35$ and individual SD 2.5, and nominal respondent counts multiplied by 1 or 0.25. Neither multiplier estimates effective sample size. Candidate packets contain 20 draws, with ordinary cost one and validation cost five. We compute initial one-step gain-per-cost priorities under four assumed bias scales; no new NHS observation or intervention is generated or observed.

Uncertainty and audit. Mean intervals use MCSE across independent replications; policy differences use their paired replication differences. They quantify Monte Carlo uncertainty conditional on the specified world or fixed record frame, not uncertainty about the NHS or BRFSS populations. Cost and budget settings share worlds and are not pooled as independent evidence. Source files have retrieval timestamps and SHA-256 records. Separate code independently checks Gaussian updates, selection cutoffs, theory, record extraction, budget compliance and hidden-target exclusion. Local protocols record the design before full execution and identify later analyses; they are not external preregistrations.

5 Results

Validation helps some Gaussian worlds and harms others. At budget 120 and validation cost 5, increasing independent offset SD from 0.15 to 0.70 changes the comparison substantially (Figure 1). At $\beta = 0.15$, two-channel KG raises regret from 0.0804 to 0.0862, a paired increase of 0.0058 [0.0042, 0.0073]. At $\beta = 0.35$ it reduces regret from 0.1541 to 0.1364; at $\beta = 0.70$ it reduces regret from 0.2503 to 0.1743, a difference of -0.0760 [$-0.0803, -0.0716$]. With no offset it chooses only ordinary measurements and matches ordinary KG. The fraction of budget spent on validation rises from 0 at zero bias to 0.916 at $\beta = 0.70$. A nonzero bias floor alone therefore does not justify an expensive finite-budget measurement policy.

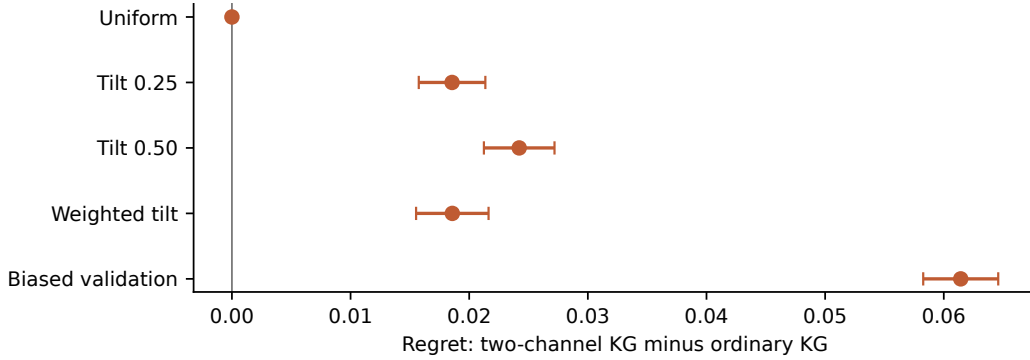


Figure 2: Paired regret differences at replay budget 144 and cost 5. Positive values mean two-channel KG is worse than ordinary KG. Intervals use paired Monte Carlo standard errors. The validation-failure condition has a separate persistent tilt; independence of its sign does not make it a valid reference channel.

The positive simulation result does not transfer automatically. Table 1 reports all ten policies at replay budget 144 and cost 5. Under tilt 0.50, ordinary KG has regret 0.1615, compared with 0.1857 for two-channel KG and 0.1884 for the balanced mixture. Two-channel KG’s paired difference from ordinary KG is $+0.0242$ $[0.0212, 0.0272]$ (Figure 2). It spends 97.2% of the budget on validation. It is close to uniform ordinary sampling (0.1848) and does not justify an overall claim of better decision efficiency. Its lower mean than the balanced mixture is a narrower same-action-set comparison. Under uniform sampling, the boundary heuristic has the lowest mean regret in this displayed grid; KG is not a universal winner.

Estimation accuracy and selection value diverge. In the tilt-0.50 replay, two-channel KG reduces target MSE from 0.02183 to 0.02129 while increasing severity regret. The weighted sensitivity shows the same direction: MSE falls from 0.02250 to 0.02194, but regret increases from 0.1773 to 0.1959. These descriptive comparisons concern different loss functions, not a causal decomposition of the failure. More precise levels across all units need not resolve the comparisons near a review boundary. Non-Gaussian observations, a working prior, myopia and unequal channel costs remain jointly present; the experiment does not identify a unique cause.

Cost, validity and model structure change the answer. For tilt-0.50 replay at budget 144, two-channel regret is 0.1626, 0.1857 and 0.1933 as validation cost rises from 2 to 5 to 10; ordinary KG remains at 0.1615. When validation itself has tilt 0.25, two-channel regret reaches 0.2226 versus ordinary KG’s 0.1612. Gaussian stress tests also reveal reversals. With overestimated bias, regret is 0.1303 versus 0.0588; with a shared group shift it is 0.1336 versus 0.0599; after target drift it is 0.1761 versus 0.1572. Heavy-tailed packet noise, unequal noise and underestimated bias yield different comparisons, all retained in the supplement. The shared-shift result is consistent with the invariant-loss argument, while also including an intentionally misspecified independent-offset acquisition model.

A larger budget can reverse the replay comparison. After the primary analysis, we specified an exploratory extension with 500 new repetitions, the unweighted tilt-0.50 mechanism, validation cost 5, and budgets 48, 144, 480 and 1,440. All four settings were fixed before this separate run.

Two-channel KG remains worse than ordinary KG at the first three budgets. At 1,440 it has regret 0.0907 versus 0.1242, a paired difference of -0.0335 $[-0.0379, -0.0291]$. It spends all additional budget on validation at the two largest budgets. Thus the last comparison concerns adaptive selection of validation units, not successful balancing of channels. Uniform validation has regret 0.1229 at the same expenditure of 1,440. The supplement retains the whole extension and its post-primary status; it does not establish an NHS spending threshold.

NHS sensitivity. The historical cohort has median nominal burnout count 3,140.5. With nominal counts treated as effective and assumed $\beta = 0$, the 20 units with greatest initial measurement value per cost all prefer ordinary packets. At assumed $\beta = 0.15$, all 20 prefer validation. With the count multiplier reduced to 0.25, only 15 prefer validation at that bias scale. This change is a sensitivity result, not an empirical estimate of nonresponse bias or a recommendation to contact those units. It demonstrates why precision and channel-validity assumptions must accompany any operational list.

6 Social-Impact Pathway and Limitations

The intended human decision is whether to spend an additional information budget before allocating scarce review capacity. The algorithm allocates a given budget among permitted measurements; it does not optimise whether to collect any new data or implement a value-based stopping rule. A usable interface would show the provisional review set, its sensitivity to precision and bias assumptions, the reason an additional observation could alter the set, expected cost and the conditions under which that channel is valid. Routine participation opportunities remain intact. Requests for immediate staff support must never wait for an acquisition algorithm to resolve a ranking.

The computational loss is deliberately limited. It values selected units equally, not employees equally, does not measure treatment benefit, and may miss a vulnerable subgroup hidden inside an organisational average. Within-group quotas preserve comparability only by assumption; they do not establish equitable access. Respondent burden, invitation effort, nonresponse, confidentiality and analysis time are absent from the packet-cost model. Twenty completed draws are purchasable by construction in the simulator, not guaranteed through real recruitment. NHS guidance emphasises connecting listening to visible action [NHS England, 2023a]; an algorithm that generates more questionnaires without follow-through could impose costs without benefit.

Before operational use, the reference measurement must be defined and tested for the same construct and population. Independent respondents alone are insufficient. A stakeholder study would need to verify the proposed review decision, acceptable delay, cost accounting and overrides. A prospective evaluation could compare equal-budget measurement policies using a prespecified independent reference and assess decision changes, burden, participation and subgroup coverage; downstream health effects would require a separate appropriate causal design. These studies have not occurred. BRFSS replay provides individual-record evidence for controlled measurement decisions, not validation of an NHS deployment.

Our theoretical formulas assume fixed Gaussian hyperparameters and independent unit states for one-coordinate acquisition. Real records violate the exact reference model; the deliberately favourable 2023 bias calibration is unavailable without extra knowledge in practice. The cohort and one held-out target year do not establish broad transfer. Replays sample with replacement from complete-case frames and exclude people with missing outcomes; weighted replay still does not resolve survey nonresponse or complex-design inference. Monte Carlo intervals do not cover these sources of uncertainty. Existing KG and MISO ideas, rather than algorithmic novelty, anchor the

contribution. The resulting evidence supports targeted scrutiny of measurement policies: repeating data can preserve bias, validation can be too costly or invalid, and reducing uncertainty is valuable only relative to the actual decision.

7 Exploratory Budget Extension

After inspecting the primary replay results, we specified a separate budget extension to test whether the short-budget comparison persists when more reference measurements are affordable. The extension protocol was recorded before its own execution, but after the primary analysis; it is therefore exploratory. It does not replace the original grid. It fixes the unweighted tilt-0.50 replay, validation cost 5, all previous-year calibration, budgets 48, 144, 480 and 1,440, and 500 repetitions with a new seed. The same unit-channel packet bank is reused across the four budgets and all five policies. All outcomes from this grid are reported, including those unfavourable to two-channel KG.

Budget	Ord. KG	Two KG	Balanced	Val. only	Uniform
48	0.1794	0.1957	0.1972	0.1986	0.1988
144	0.1593	0.1867	0.1897	0.1882	0.1815
480	0.1367	0.1500	0.1800	0.1635	0.1605
1440	0.1242	0.0907	0.1499	0.1229	0.1380

Table 2: Exploratory budget extension: mean severity regret, 500 paired repetitions at each budget. This is a separate run from the 1,000-repetition primary grid. The complete CSV includes actual spending and paired Monte Carlo intervals.

Two-channel KG remains worse than ordinary KG at budgets 48, 144 and 480. At budget 1,440, the comparison reverses: regret is 0.09070 versus 0.12422, a paired difference of -0.03352 with 95% Monte Carlo interval $[-0.03793, -0.02910]$. This is evidence of a budget-dependent comparison within the specified replay, not a measured NHS cost threshold. At budgets 480 and 1,440, two-channel KG spends all additional resources on validation. The improvement must not be described as successful balancing of two channels. At 1,440, uniform validation also uses only validation, spends the same budget, and has regret 0.12288; the comparison with KG concerns which units receive those reference packets. The budget is 30 ordinary-packet cost equivalents per area, or 288 validation packets in total at cost 5, on top of the shared initial observations. A real price per completed packet is not inferred.

8 Estimation, Cost and Misspecification Diagnostics

The MSE/regret comparison makes the target dependence of acquisition tangible. MSE averages squared level errors across every unit; severity regret evaluates only the selected set relative to the oracle with the same quota. A small reduction in aggregate MSE can coexist with a less accurate ordering around the selection boundary. The replay comparisons below are descriptive effects of full policies. They do not isolate how much of the difference is due to Gaussian approximation, prior calibration, per-packet noise, cost or myopic action choice.

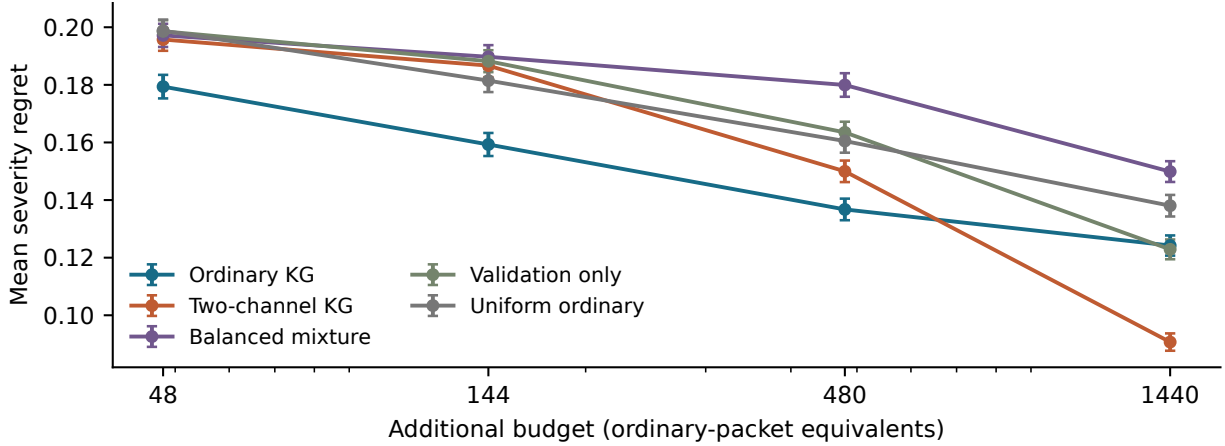


Figure 3: Exploratory budget sensitivity on one controlled replay mechanism. Error bars are 95% Monte Carlo intervals. All four budgets were specified before running this extension. Ordinary-only policies cannot access validation; the balanced mixture and validation-only policy provide comparisons with that access.

Scenario	Ord. loss	Two loss	Ord. MSE	Two MSE	Val. cost
Uniform	0.1124	0.1124	0.01802	0.01802	0.0%
Tilt 0.25	0.1570	0.1755	0.02116	0.02092	90.2%
Tilt 0.50	0.1615	0.1857	0.02183	0.02129	97.2%
Weighted tilt	0.1773	0.1959	0.02250	0.02194	97.2%
Biased validation	0.1612	0.2226	0.02182	0.02308	97.2%

Table 3: Primary replay at budget 144 and validation cost 5. “Ord.” and “Two” refer to ordinary-only and two-channel KG with matched bias-aware inference. Validation cost is the fraction of the two-channel policy’s budget spent on that channel. Target MSE and selection loss are distinct.

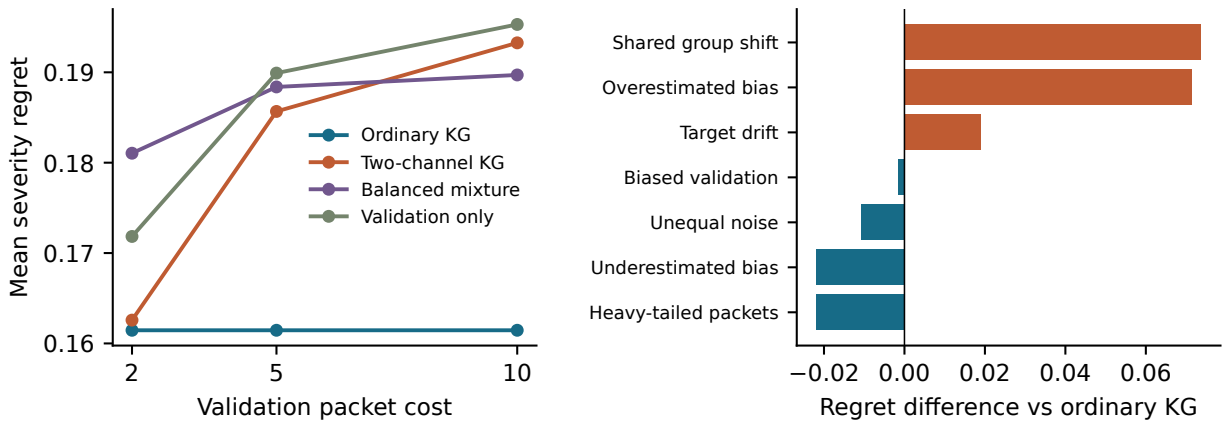


Figure 4: Left: validation-cost sensitivity on tilt-0.50 replay at budget 144. Right: synthetic stress-world regret differences at budget 120 and cost 5; positive values favour ordinary KG. All stress results remain in the numerical release, including small or adverse comparisons.

9 Research Transparency

This study uses public aggregate NHS data and public-use BRFSS records. It conducts no new recruitment, intervention or private-data linkage. The computational benchmark does not attribute individuals’ mental health to their jobs and does not diagnose a clinical condition. Source: CDC BRFSS. This independent research and its conclusions are not endorsed by CDC, HHS, the United States Government or NHS organisations. Existing source terms continue to govern original materials; a licence for newly authored code does not relicense agency materials.

AI assistance was used extensively for literature discovery, mathematical derivation, code generation, numerical auditing, analysis and manuscript preparation. Automated checks and independent implementations are documented; they do not replace the named author’s responsibility for reviewing the manuscript, verifying references, understanding the analysis and approving any submission. No external stakeholder co-design, ethics approval, funding award or deployment is claimed. No real institutional allocation has been made from the research. Authorship and any necessary institutional declarations must accurately reflect the actual work and affiliation when submitted.

The source package contains runnable experiment scripts, derived public-data frames, complete summary CSVs, a results workbook, acquisition/source manifests, numerical-verification scripts, local analysis protocols and source snapshots for the primary runs. Detailed simulation metrics can be regenerated with the stored seeds. The separate exploratory extension also retains replication-level arrays. Working-directory paths and personal identifiers are excluded from the anonymous PDF text. Public-data access, re-download and source validation depend on continued availability of the cited official releases; included derived-data reproduction does not require private credentials or a GPU.

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A Notation and Complete Reference-Model Results

The propositions below restate the model self-containedly. To distinguish local notation from channel cost, the off-diagonal covariance is denoted c_i in these propositions (the main paper calls it u_i). Channel cost is written explicitly as cost_{id} . The results concern the declared independent-unit Gaussian model with fixed hyperparameters. They do not infer real survey bias from aggregate scores.

B Decision value of another measurement

Let $\mathcal{S} = \{S : |S \cap g| = k_g \text{ for every group } g\}$, $K = \sum_g k_g > 0$, and let lower θ_i mean greater need. We use

$$L(S, \theta) = \frac{1}{K} \left\{ \sum_{i \in S} \theta_i - \min_{A \in \mathcal{S}} \sum_{i \in A} \theta_i \right\}. \quad (8)$$

Thus review slots have equal weight across all selected units; this is not equal weighting of group-specific average losses. The Bayes decision selects the k_g smallest posterior target means within each group. Linearity of expectation suffices for this decision, even with dependent targets. Neither this loss nor its minimizer is a probability-of-exact-set-recovery objective. Fixed-budget subset selection and boundary-directed acquisition have substantial prior art [Chen et al., 2008, Kalyanakrishnan et al., 2012, Chen et al., 2014].

For the acquisition calculation, assume independent unit states, fixed hyperparameters, and a Gaussian posterior $(\theta_i, b_i) \mid D \sim N((m_i, a_i)^\top, P_i)$, where $P_i = \begin{pmatrix} v_i & c_i \\ c_i & q_i \end{pmatrix}$. An ordinary packet measures $\theta_i + b_i$; an ideal validation packet measures θ_i . Their loadings are $h_R = (1, 1)^\top$ and $h_A = (1, 0)^\top$. A packet has conditionally independent Gaussian noise variance $r_{id} > 0$. Its posterior-mean innovation has standard deviation

$$s_{id} = \frac{|(P_i h_d)_1|}{\sqrt{h_d^\top P_i h_d + r_{id}}}. \quad (9)$$

For a independent respondent draws with individual noise variance r , the packet mean has variance r/a ; a persistent offset does not shrink this way. Conditional unbiasedness of the validation channel is a reference-model assumption, not a property inferred from the NHS data.

Proposition 1 (One-step severity value). *For an action affecting only unit i in a group with $0 < k_g < |g|$, let t_i be the k_g th smallest posterior mean among the other units in that group and set $\Delta_i = |m_i - t_i|$. Its expected reduction in posterior Bayes risk is*

$$\text{KG}_{id} = \frac{1}{K} \{s_{id}\varphi(\Delta_i/s_{id}) - \Delta_i\Phi(-\Delta_i/s_{id})\}, \quad (10)$$

with value zero at $s_{id} = 0$. Groups selecting none or all of their units have zero local acquisition value. For finite gaps and positive s , the value strictly increases with s and decreases with the gap.

This is a closed-form specialization of Gaussian knowledge gradient [Frazier et al., 2008, 2009], closely related to multi-information-source optimization [Poloczek et al., 2017]; it is not a new subset-selection algorithm. Our policy compares affordable actions by $\text{KG}_{id}/\text{cost}_{id}$. The common $1/K$ factor can be omitted when choosing the action. This myopic cost ratio is not globally optimal for a fixed multistep budget; Appendix C gives a counterexample, consistent with known nonconcavity of information value [Frazier and Powell, 2010]. If an observation updates several units, Equation (10) must be replaced by the joint decision value.

Proposition 2 (Limits of repeated ordinary measurement). *Suppose $\theta \sim N(\mu, \tau^2)$ and $b \sim N(0, \beta^2)$ are independent, with $\tau, \beta > 0$, and $Y_j = \theta + b + \epsilon_j$ has independent $\epsilon_j \sim N(0, r)$. Then*

$$\text{Var}(\theta \mid Y_{1:n}) = \tau^2 - \frac{\tau^4}{\tau^2 + \beta^2 + r/n} \rightarrow \frac{\tau^2\beta^2}{\tau^2 + \beta^2}. \quad (11)$$

For two independent units with this common prior and one review slot, even exactly observing both $\theta_i + b_i$ leaves Bayes severity regret

$$R_\infty = \frac{\tau}{\sqrt{\pi}} \left(1 - \frac{\tau}{\sqrt{\tau^2 + \beta^2}}\right) > 0. \quad (12)$$

This is a lower bound for every policy restricted to noisy ordinary measurements under that prior, including adaptive policies.

The explicit selection-risk expression concerns independent unit-specific offsets. Partial identification despite unlimited samples is a broader, established phenomenon [Altschuler et al., 2019]; the next result shows why level uncertainty alone does not establish selection harm.

Proposition 3 (Shared group shifts do not change this loss). *For arbitrary constants u_g , define $\theta'_i = \theta_i + u_g$ for $i \in g$. For every feasible S , $L(S, \theta') = L(S, \theta)$, and the collection of optimal feasible sets is unchanged. Consequently, if the only persistent ordinary-channel offset is common within each group, exact ordinary means identify an optimal within-group selection even when target levels remain unidentified.*

Thus a useful measurement reduces uncertainty about relevant differences, not necessarily about absolute levels. A policy that models common offsets as independent unit offsets can value validation for distinctions that are irrelevant to fixed within-group slots.

Proposition 4 (When a second channel resolves uncertainty). *In the independent-unit reference model, ideal validation has greater one-step value than an ordinary packet at the same unit and equal cost if and only if*

$$\frac{v_i^2}{v_i + r_{iA}} > \frac{(v_i + c_i)^2}{v_i + q_i + 2c_i + r_{iR}}, \quad (13)$$

provided the group has a nontrivial quota. If $c_i = 0$ and $v_i > 0$, this reduces to $r_{iA} < q_i + r_{iR}$. If validation instead measures $\theta + d$ with a second persistent offset, independent priors with positive variances $\tau^2, \beta^2, \delta^2$ imply

$$\text{Var}(\theta \mid \theta + b, \theta + d) = (\tau^{-2} + \beta^{-2} + \delta^{-2})^{-1} > 0. \quad (14)$$

Independence between channel offsets therefore does not suffice to make validation unbiased for an individual unit.

For unequal costs, the comparison uses the full cost-normalized expression in Equation (10), not merely variance reduction per cost. Cheap biased sources versus expensive reference sources are already central to multi-fidelity identification [Poiani et al., 2024] and selective proxy auditing [Ao et al., 2026]. These propositions characterize which parts of that information problem matter for this decision; they do not identify unknown survey bias or provide guarantees under misspecified channel models.

C Proofs and an allocation counterexample

Bayes decision and Gaussian update. Conditional on D , the oracle term in Equation (8) is independent of the chosen set. Thus minimizing expected loss is equivalent to minimizing $\sum_{i \in S} m_i$ subject to the fixed quotas. A deterministic index rule may resolve equal means without changing the objective. For a Gaussian linear observation $W = h^\top(\theta, b) + \epsilon$, write $V = h^\top P h + r > 0$. Conditioning yields

$$P^+ = P - P h h^\top P / V, \quad m^+ = m + (P h)_1 \{W - h^\top(m, a)\} / V.$$

The predictive residual is $N(0, V)$, which proves Equation (9). This calculation conditions on fixed hyperparameters; learning a shared hyperparameter from the new observation can also update other units and invalidate the one-coordinate simplification.

Proof of Proposition 1. Order the other units' means as $a_{(1)} \leq \dots \leq a_{(|g|-1)}$. With $t_i = a_{(k_g)}$, the optimal posterior-mean sum in this group equals $\sum_{j < k_g} a_{(j)} + \min(m_i, t_i)$. After the observation, replace m_i by $m_i + sZ$, where $Z \sim N(0, 1)$; all other group contributions are unchanged. The tower property cancels the expected oracle-loss term from the before/after Bayes risk. Therefore the value is

$$K^{-1} \{\min(m_i, t_i) - \mathbb{E} \min(m_i + sZ, t_i)\} = K^{-1} \mathbb{E}(sZ - |m_i - t_i|)_+.$$

Integrating the Gaussian tail gives Equation (10). For $s > 0$ and $\Delta \geq 0$, differentiation gives $\partial(K \text{KG})/\partial s = \varphi(\Delta/s) > 0$ and $\partial(K \text{KG})/\partial \Delta = -\Phi(-\Delta/s) < 0$. At $s = 0$, the mean does not change and the value is zero. For a quota of zero or the entire group, the feasible choice in that group is fixed; the tower property gives zero expected improvement. Equal means cause no problem because the optimum sum, rather than a discontinuous membership indicator, is the value function. $\square$

Proof of Proposition 2. The sample mean is sufficient and has variance $\tau^2 + \beta^2 + r/n$, with covariance τ^2 with θ . Gaussian conditioning proves Equation (11). For two units, set $T_i = \theta_i + b_i$. Their conditional means $m_i = \mu + \lambda(T_i - \mu)$, with $\lambda = \tau^2/(\tau^2 + \beta^2)$, are independent normals with standard deviation $\tau^2/\sqrt{\tau^2 + \beta^2}$. The minimum of two independent normals of common mean μ and standard deviation s has expectation $\mu - s/\sqrt{\pi}$. Hence

$$\mathbb{E} \min(m_1, m_2) - \mathbb{E} \min(\theta_1, \theta_2) = \frac{\tau}{\sqrt{\pi}} - \frac{\tau^2}{\sqrt{\pi(\tau^2 + \beta^2)}},$$

which is Equation (12). Any adaptive sequence of ordinary observations can be simulated from exact (T_1, T_2) and independent observation and policy randomness. Exact T therefore contains at least as much decision-relevant information as that sequence; its optimal Bayes risk is a lower bound for all such policies. $\square$

A current-posterior version of the variance floor. For arbitrary positive-semidefinite P with entries (v, c, q) , exact knowledge of $T = \theta + b$ leaves $v - (v + c)^2 / (v + q + 2c)$ when $\text{Var}(T) > 0$. If $\text{Var}(T) = 0$, T is already known and its acquisition does not change v ; positive semidefiniteness implies $v + c = 0$. After learning T exactly, $\text{Cov}(\theta, T \mid T) = 0$, so further ordinary sampling has zero target-mean innovation. Under independent units, positive remaining v , a nontrivial quota, and finite noise, ideal validation still has strictly positive one-step value. Affordability and the amount of remaining budget are separate constraints.

Proof of Proposition 3. Every feasible set has the same additional sum $\sum_g k_g u_g$ under the transformation. It appears both in the selected-set sum and the minimized oracle sum, so it cancels. All comparisons among feasible sums are also unchanged. If ordinary means equal $\theta_i + b_g$, this same argument applies with $u_g = b_g$ to recover the optimal feasible sets. $\square$

Why marginal uncertainty can be irrelevant. For two units with $\theta_1 = \theta_2 = Z$, $Z \sim N(0, 1)$, and one slot, the selection regret is always zero. An exact observation of either unit updates both posterior means to Z , so the true joint information value is zero. Updating only one coordinate incorrectly would give $\varphi(0) = 1/\sqrt{2\pi}$. This shows why the independence/one-coordinate assumption of Proposition 1 matters when offsets or targets have shared structure.

Proof of Proposition 4. At the same unit the gap is the same for both actions, and Proposition 1 is strictly increasing in s . Substituting h_A and h_R into Equation (9) and comparing squared innovations gives Equation (13). For $c = 0$ and $v > 0$, the equal numerators v^2 cancel. For the final statement, conditional on θ , exact $T = \theta + b$ and $U = \theta + d$ have independent likelihoods with variances β^2 and δ^2 . Adding their precisions to the prior precision proves Equation (14). Repeated independent observation noise can reveal T and U , but it does not remove their persistent offsets. If both channels instead share the same offset, their exact values supply only one such equation. $\square$

A fixed-budget counterexample to the cost ratio. There is one known alternative of value zero and one target $\theta \sim N(0, 1)$, with one slot and budget two. A cheap direct measurement of θ has noise variance one and cost one. A more accurate direct measurement has noise variance 0.1 and cost two. Both have zero initial gap. Their initial knowledge gradients are $1/\sqrt{4\pi} = 0.282095$ and $1/\sqrt{2.2\pi} = 0.380377$, respectively. Their cost ratios therefore favor the cheap action: $0.282095 > 0.190188$. Only another cheap action is affordable afterward. The two cheap measurements produce posterior mean variance $2/3$, hence total expected improvement $\sqrt{2/3}/\sqrt{2\pi} = 0.325735$. Spending the budget on the more accurate action yields 0.380377, a larger value. The policy can fail even with correct Gaussian beliefs and conditionally unbiased channels; one-step value per cost is not the multistep budget objective.

D Algorithms and Exact Computation

D.1 Policy state and update

Each replication starts with a two-coordinate state per unit. Its state stores target mean, offset mean, target variance, offset variance, their covariance, and ordinary/validation draw counts. For each unit i and affordable channel d , compute the relevant other-unit quota boundary t_i , the innovation SD and Equation (10). Choose the action with maximum gain per cost, read the next unused entry in that unit-channel observation stream, apply Gaussian conditioning, and deduct the channel cost. Ties use deterministic flattened unit/channel order for KG-based rules. Recompute boundaries after every observation. The final review set consists of the smallest posterior means within each group, with stable index ties.

One update has constant cost per affected unit. Computing all boundaries by sorting costs $O(M \log M)$ per replication in this implementation; evaluating $2M$ action values is linear. The implementation is vectorised over independent replications. It does not claim an optimal computational complexity or a new data structure. Finite-precision normal-tail calculations are clamped to nonnegative values to avoid tiny cancellation artefacts. A budget is integer-valued and checked at every step. Validation-only policies may leave a remainder smaller than the validation cost. Comparisons report both available budget and actual spending.

D.2 Ten policy definitions

1. Uniform ordinary: choose a unit uniformly and acquire an ordinary packet, retaining the common bias-aware posterior.
2. Uniform, bias blind: the same action rule with assumed bias mean and variance zero.
3. Maximum variance: acquire an ordinary packet at the unit with largest target posterior variance.
4. Boundary heuristic: acquire ordinary data at the largest $2\sqrt{v_i} - |m_i - t_i|$. This fixed-budget heuristic is not the LUCB algorithm.
5. Ordinary KG: maximise one-step severity gain using only ordinary packets.
6. Bias-blind KG: ordinary KG with zero assumed offset.
7. Equal-action mixture: sample uniformly over affordable unit-channel pairs. Equal action probability does not mean equal expenditure.
8. Balanced mixture: before affordability effects, choose validation with probability $1/(1 + c_A)$ and a uniform unit; otherwise ordinary. If a requested validation packet is unaffordable, use an ordinary packet. This balances expected expenditure before terminal effects.
9. Validation only: choose a uniform unit and use only validation after the common initial ordinary phase. Stop when no validation packet is affordable.
10. Two-channel KG: maximise one-step severity gain per cost over both affordable channels, using the same posterior update as the bias-aware baselines.

The full-code release also contains a dormant terminal-rule variant used for implementation exploration; no reported experiment uses it. Shared inference distinguishes action selection from the

benefits of assuming a bias model. Common channel access distinguishes two-channel acquisition efficiency from the value of access to a different measurement source.

D.3 Pairing, estimation and stopping

Each world or replay scenario generates a packet bank indexed by replication, unit, channel and within-channel packet count. Policies observing the same count for the same unit-channel pair see the same packet, regardless of their elapsed action number. Latent worlds and banks are shared across cost and budget settings within a primary scenario. Independent policies may take different numbers of packets, but do not overwrite or consume another policy’s bank. Randomised baselines have explicit seeds.

For R independent replications and a metric z_r , the stored MCSE is $s(z)/\sqrt{R}$ and the displayed interval is $\bar{z} \pm 1.96$ MCSE. For two policies, compute the differences $z_r^{(1)} - z_r^{(0)}$ first and apply the same calculation; do not combine their marginal standard errors as if they were independent. Primary CSVs store paired regret and overlap comparisons with ordinary KG and bias-aware uniform ordinary. The exploratory extension retains replication-level arrays to verify these computations. No cross-scenario pooled significance test is reported, and simultaneous coverage over the full grid is not claimed.

In finite-file replay, a *fixed-length primitive packet mean* from iid uniform record draws is unbiased for its declared finite-frame mean. Under weight-proportional sampling it is unbiased for the constructed weight-proportional distribution. This statement does not make the final posterior estimate, or a unit’s empirical mean after outcome-adaptive sampling, unbiased. Counts depend on observed outcomes. Interval coverage is calculated against each fixed-frame target across replay repetitions; it does not establish design-based population coverage.

E Synthetic Worlds in Full

Targets are independent $N(5, 0.35^2)$ for 40 units, arranged in four groups of ten with quota two. In the base worlds, an offset $b_i \sim N(0, \beta^2)$ persists across initial and subsequent ordinary packets. Validation has fresh noise and no persistent offset. Initial sample size and additional packet size both equal 20. Conditional packet-noise SD is $2.5/\sqrt{20}$. Budgets are 40 and 120, validation costs 2, 5 and 10, and ordinary cost one. Four bias scales and six cost/budget pairs yield 24 settings and 240 policy rows. Each of the four worlds has 2,000 independent replications; the 24 settings are not 48,000 independent latent panels.

Seven stress worlds add 70 policy rows, each with 2,000 independent replications, budget 120 and validation cost 5. Underestimated bias uses true SD 0.70 and assumed SD 0.15; overestimated bias uses true SD zero and assumed SD 0.70. The validation-bias world adds an independent persistent validation offset with SD 0.35 while the update treats validation as unbiased. The heavy-tail world replaces Gaussian packet noise with Student- t_3 noise scaled to the same packet variance. It draws heavy-tailed *packet* means directly; a packet is not implemented as an average of 20 individual heavy-tailed observations. Unequal noise multiplies each unit’s noise SD by a lognormal factor with log SD 0.4 while inference keeps the common SD. Target drift adds independent $N(0, 0.35^2)$ changes after initial observations; all later observations measure the changed target while inference assumes a static state. Shared-group bias uses a single SD-0.70 offset per group, but inference treats offsets as independent. Unless specified otherwise, true and assumed bias SDs are 0.35.

All seven stresses are deliberately misspecified relative to the exact propositions. The purpose is to locate failures, not to claim robustness from passing a chosen subset. Uniform/blind baselines

use the same observation law as other policies within a world. Unit-mean target MSE and interval coverage concern the post-drift target where applicable. Unbounded Gaussian targets are a controlled mathematical world, not a generative model claiming that NHS scores can lie outside their scale.

F Public Microdata Construction and Temporal Separation

F.1 Source and cleaning

The downloaded 2023 BRFSS ASCII file contains 433,323 records; the 2024 file contains 457,670. Official SAS import programs and variable-layout documentation determine required ASCII offsets. Reading streams from ZIP members avoids expanding gigabyte-size text files. The required variables are state, core mental-health days, employment, final combined survey weight, stratum, PSU and the supplied mental-health category used for validation. Core outcome codes 1–30 and 88 are valid; 77, 99 and blanks are excluded. Employment codes 1 and 2 denote employed for wages and self-employed. The score is a transparent rescaling, not a newly validated wellbeing questionnaire or a measure of job-caused distress.

Cleaning stage	2023	2024
Complete public file	433,323	457,670
Employed/self-employed before outcome filtering	215,794	225,524
Employed with valid outcome, all areas	212,989	222,734
Common-state/DC complete-outcome frame	206,098	214,151
Number of common areas	48	48
Minimum records per included area	1,247	1,232
Maximum records per included area	13,474	22,708

Table 4: Executed BRFSS cohort construction. The cross-year intersection excludes territories, Kentucky and Pennsylvania (absent from 2023), and Tennessee (absent from 2024). Annual files are independent cross-sections, not a person panel.

All 420,249 cleaned primary records were independently compared with the original ASCII slices. All 457,670 original 2024 records were also checked against the official SAS-transport file: categorical/design fields agree, and the maximum absolute weight discrepancy is 4.997×10^{-5} from the ASCII representation’s decimal precision. Four employed Illinois records have a core mental-health response inconsistent with the supplied categorical derivative. The same discrepancy exists in both original formats. We keep the core responses, record the discrepancy and do not selectively correct or exclude those records. This check demonstrates why matching aggregate codebook frequencies alone is insufficient.

The actual compressed ASCII sizes were 60,057,185 and 53,411,907 bytes. The acquired 2024 text has 2,061-character records, despite a stale longer-file description on the annual webpage. Source offsets are verified against the actual import programs and layout entries; page descriptions do not override measured files. Retrieval timestamps, resolved URLs, byte counts, SHA-256 hashes and ZIP member CRCs identify the downloaded releases. These are locally computed identity checks, not publisher-supplied hashes or claims of access to the first published vintage.

F.2 Targets and the sampling mechanism

For area i and year y , let $p_{iy}(x)$ be the empirical probability of score x among complete employed records. The primary target is $\theta_{i,2024} = \sum_x p_{i,2024}(x)x$. All 2024 means are reserved for final evaluation. A fixed 2023 common target prior uses $\mu = 8.55693$ and $\tau = 0.13882$, calculated from its 48 area means. The 2023 within-area variance supplies packet-noise calibration. No held-out 2024 full-frame moment is given to the action selector.

For a random sign $s_i \in \{-1, +1\}$ fixed throughout a replication, the ordinary sampler uses

$$p_{i,2024}^{(s_i)}(x) = \frac{p_{i,2024}(x) \exp\{s_i \kappa(x - \mu_{i,2023})/\sigma_{i,2023}\}}{\sum_z p_{i,2024}(z) \exp\{s_i \kappa(z - \mu_{i,2023})/\sigma_{i,2023}\}}.$$

Centering cancels from the normalisation; the scale remains frozen to 2023. The same plus/minus construction on 2023 gives two hypothetical mean offsets. Their midpoint is the assumed bias mean; their population SD is the assumed bias SD. The average of the two within-law variances supplies ordinary noise. This centre is not generally zero because the outcome distribution is asymmetric. For $\kappa = 0.25$ and 0.50 , the average calibrated bias SDs are approximately 0.6865 and 1.4952 , respectively. These are deliberately strong controlled tilts relative to the between-area spread, not plausible-bias estimates derived from actual recruitment validation.

Validation draws uniformly from the hidden empirical distribution except in its separate failure scenario. There it has an independent persistent sign and tilt 0.25 , yet the update continues to assume an unbiased channel. The exact two-sign mechanism is not fitted as a discrete posterior; the policy deliberately uses the same moment-matched Gaussian approximation. Its calibration is favourable because it knows a hypothetical sampling-law family, but it is not an oracle using realised 2024 offsets or reference means.

Weighted sensitivity replaces p_{iy} by the empirical probability proportional to each record's final survey weight. Its target is $\sum_j w_j x_j / \sum_j w_j$ in the retained finite file. The common 2023 prior has $\mu = 8.41342$ and $\tau = 0.14436$. Sampling in proportion to full-frame weights is an additional offline assumption. Retaining stratum and PSU information does not turn this replay into complex-survey population inference. A Kish weight-concentration count, when reported in source diagnostics, is not an effective sample size accounting for clustering, stratification or outcome-specific design effects.

F.3 What is independently observed

Only the original recorded outcomes, employment categories, geography and design fields are observed. The additional recruitment laws, channel validity, signs, packet costs, repeated sampling and review capacities are constructed. Full-file means are evaluation references within those laws. The replay is therefore stronger than a Gaussian-only simulation in outcome realism but weaker than a prospective measurement experiment in recruitment validity. The NHS and BRFSS sources answer different questions and are never pooled as if they measured the same people or construct.

G NHS Illustration and Proposed Use

The retained historical cohort has 190 complete-five-year trust reporting units and 189 underlying legal entities; two reporting sectors belong to one legal entity. Group sizes are 110 acute/acute-community, 43 mental-health/learning-disability and related, 13 community, 13 acute specialist and 11 ambulance units. Applying $k_g = \lfloor 0.2|g| \rfloor$ gives 22, 8, 2, 2 and 2 slots, totalling 36. These are modelled review capacities, not existing NHS allocation rules. The complete-five-year restriction

defines an illustration cohort; it does not imply all NHS trusts were stable or comparable throughout the period.

For each assumed bias scale $\{0, 0.15, 0.35, 0.70\}$ and precision multiplier $\{1, 0.25\}$, initialise a Gaussian prior with group-mean plug-in centre and $\tau = 0.35$, then condition on observed 2025 burnout scores using variance $2.5^2/(n_i f)$. Here n_i is the reported nominal composite response count and f is a sensitivity multiplier. Neither 2.5 nor f is estimated from respondent microdata. The 2025 group-mean plug-in also means this is an illustrative conditional state, not a fully integrated empirical-Bayes uncertainty analysis. Candidate packets have size 20, assumed individual SD 2.5, ordinary cost one and validation cost five. No future outcomes are drawn: the output is an initial one-step value table under stated assumptions.

At $f = 1$, average posterior SD rises from approximately 0.0477 with zero assumed bias to 0.2488 at bias SD 0.35; the latter is close to the repeat-only floor SD 0.2475. This does not empirically establish that NHS nonresponse bias is 0.35, nor that a validation packet of that cost exists. It indicates which unverifiable assumptions drive the apparent preference for a new channel. The machine-readable output labels every row as a model-conditional scenario with no new NHS observations.

A possible operational sequence is: retain routine access to staff listening; agree the review decision and comparison groups; define the target construct; assess whether any alternative measurement can credibly target it; account for invitation, response, completion and analysis costs; inspect how the provisional set changes under plausible assumptions; and only then consider additional acquisition. Staff representatives and organisational users would need to review burdens, confidentiality and overrides. This sequence is a proposed study design, not a workflow validated with NHS staff. Thresholds suppressing small survey counts protect confidentiality and must not be treated as sufficient sample-size criteria for accurate ranking. Quarterly engagement scores and the annual burnout composite must not be substituted without a measurement argument.

H Independent Verification and Reproduction

The theoretical verifier compares 60 closed-form Gaussian gains with numerical quadrature (maximum error below 5×10^{-16}), checks 200 shared-shift invariance cases, and verifies the two-unit irreducible Bayes risk using 400,000 Monte Carlo draws. The exact risk is 0.0475387; the simulation estimate is 0.0477082 with MCSE 0.0001909. A separate allocation verifier includes unequal group sizes, stable ties and 120 general covariance updates. The implementation audit uses 700 successive full-matrix updates, 80 numerical expected-top- k integrations, small-seed step-by-step comparisons of all ten policies, and independent recomputation of 240 replay calibration rows.

Four complete saved primary result rows were independently reproduced with full-matrix posterior arithmetic, two using 2,000 synthetic replications and two using 1,000 replay replications. Tiny differences in arithmetic order can choose another mathematically tied acquisition: six synthetic paths exhibited near-tied actions whose maximum independent value difference was 1.73×10^{-18} . Re-evaluation on the executed paths reproduces saved metrics. Perturbing the hidden 2024 reference metadata changes evaluation MSE but leaves all ten policies' streams, actions and posteriors unchanged. This is a targeted leakage check plus code inspection, not a proof against every possible implementation error.

The numerical environment uses Python 3.13.1, NumPy 2.5.3, pandas 3.0.5, SciPy 1.18.1 and Matplotlib 3.11.1. Primary synthetic and replay runs took approximately 95 and 74 seconds, respectively, in this local CPU environment. The exploratory budget extension took 45 seconds with measured peak resident memory approximately 1.08 GB. These are single-run execution mea-

surements, not portable performance guarantees. No GPU, model training or private service was used.

The repository includes environment requirements, scripts for verified download and parsing, manifest hashes, exact random seeds, summary results and paper-generation commands. Run the primary simulations, primary replay, NHS scenarios, exploratory extension and independent verifiers separately; the report script regenerates tables and figures from CSVs. Changes to upstream releases may require retrieval of the recorded version rather than silently accepting new bytes. Primary source snapshots preserve code and protocol versions at execution; later report or audit scripts do not retroactively preregister a result.

I Complete Primary Mean-Regret Grid

The following compact tables contain every declared primary policy-setting mean regret: 240 Gaussian, 70 stress and 180 replay rows. Full precision, MCSE, paired comparisons, overlap, MSE, coverage, spending and allocation counts are in the accompanying CSVs and workbook. Table abbreviations are U (uniform ordinary), UB (uniform bias blind), V (maximum variance), B (boundary heuristic), O (ordinary KG), OB (bias-blind ordinary KG), E (equal-action mixture), M (cost-balanced mixture), A (validation-only uniform), and T (two-channel KG). Lower values are better. No ranking is inferred across different worlds or estimands.

Table 5: Gaussian: all ten policy mean regrets. Full precision and uncertainty are in the CSV release.

Setting	U	UB	V	B	O	OB	E	M	A	T
b=0.00, c=2, B=40	0.1580	0.1580	0.1494	0.1094	0.1101	0.1101	0.1734	0.1687	0.1789	0.1101
b=0.00, c=2, B=120	0.1063	0.1063	0.0971	0.0584	0.0584	0.0584	0.1267	0.1209	0.1420	0.0584
b=0.00, c=5, B=40	0.1580	0.1580	0.1494	0.1094	0.1101	0.1101	0.1869	0.1729	0.1962	0.1101
b=0.00, c=5, B=120	0.1063	0.1063	0.0971	0.0584	0.0584	0.0584	0.1584	0.1341	0.1738	0.0584
b=0.00, c=10, B=40	0.1580	0.1580	0.1494	0.1094	0.1101	0.1101	0.1880	0.1754	0.2024	0.1101
b=0.00, c=10, B=120	0.1063	0.1063	0.0971	0.0584	0.0584	0.0584	0.1736	0.1350	0.1909	0.0584
b=0.15, c=2, B=40	0.1684	0.1686	0.1619	0.1250	0.1242	0.1245	0.1783	0.1790	0.1855	0.1252
b=0.15, c=2, B=120	0.1234	0.1234	0.1148	0.0822	0.0804	0.0808	0.1347	0.1317	0.1437	0.0796
b=0.15, c=5, B=40	0.1684	0.1686	0.1619	0.1250	0.1242	0.1245	0.1935	0.1829	0.2033	0.1241
b=0.15, c=5, B=120	0.1234	0.1234	0.1148	0.0822	0.0804	0.0808	0.1664	0.1416	0.1801	0.0862
b=0.15, c=10, B=40	0.1684	0.1686	0.1619	0.1250	0.1242	0.1245	0.1996	0.1845	0.2099	0.1242
b=0.15, c=10, B=120	0.1234	0.1234	0.1148	0.0822	0.0804	0.0808	0.1796	0.1491	0.1962	0.0860
b=0.35, c=2, B=40	0.2069	0.2074	0.1977	0.1769	0.1751	0.1753	0.2001	0.2011	0.1980	0.1567
b=0.35, c=2, B=120	0.1772	0.1779	0.1720	0.1604	0.1541	0.1540	0.1576	0.1604	0.1528	0.0982
b=0.35, c=5, B=40	0.2069	0.2074	0.1977	0.1769	0.1751	0.1753	0.2141	0.2088	0.2172	0.1762
b=0.35, c=5, B=120	0.1772	0.1779	0.1720	0.1604	0.1541	0.1540	0.1886	0.1805	0.1931	0.1364
b=0.35, c=10, B=40	0.2069	0.2074	0.1977	0.1769	0.1751	0.1753	0.2193	0.2139	0.2239	0.1765
b=0.35, c=10, B=120	0.1772	0.1779	0.1720	0.1604	0.1541	0.1540	0.2017	0.1850	0.2108	0.1510
b=0.70, c=2, B=40	0.2683	0.2705	0.2651	0.2613	0.2594	0.2587	0.2411	0.2463	0.2313	0.1841
b=0.70, c=2, B=120	0.2605	0.2621	0.2573	0.2580	0.2503	0.2509	0.1912	0.2048	0.1666	0.1095
b=0.70, c=5, B=40	0.2683	0.2705	0.2651	0.2613	0.2594	0.2587	0.2592	0.2636	0.2585	0.2341
b=0.70, c=5, B=120	0.2605	0.2621	0.2573	0.2580	0.2503	0.2509	0.2250	0.2375	0.2241	0.1743
b=0.70, c=10, B=40	0.2683	0.2705	0.2651	0.2613	0.2594	0.2587	0.2666	0.2683	0.2701	0.2508
b=0.70, c=10, B=120	0.2605	0.2621	0.2573	0.2580	0.2503	0.2509	0.2473	0.2537	0.2482	0.2139

Table 6: Stress: all ten policy mean regrets. Full precision and uncertainty are in the CSV release.

Setting	U	UB	V	B	O	OB	E	M	A	T
underestimated bias	0.2637	0.2635	0.2601	0.2550	0.2531	0.2528	0.2447	0.2542	0.2366	0.2312
overestimated bias	0.1089	0.1056	0.0951	0.0955	0.0588	0.0574	0.1803	0.1555	0.1900	0.1303
biased validation	0.1732	0.1741	0.1693	0.1547	0.1473	0.1480	0.1935	0.1838	0.2062	0.1458
heavy tails	0.1716	0.1732	0.1672	0.1576	0.1493	0.1483	0.1744	0.1726	0.1769	0.1273
target drift	0.2113	0.2121	0.1931	0.1945	0.1572	0.1613	0.2697	0.2396	0.3040	0.1761
heteroscedastic	0.1857	0.1866	0.1782	0.1728	0.1609	0.1600	0.2001	0.1897	0.2077	0.1500
shared group bias	0.1108	0.1178	0.0977	0.1017	0.0599	0.0704	0.1814	0.1605	0.1906	0.1336

Table 7: Replay: all ten policy mean regrets. Full precision and uncertainty are in the CSV release.

Setting	U	UB	V	B	O	OB	E	M	A	T
Uniform, c=2, B=48	0.1691	0.1691	0.1679	0.1434	0.1476	0.1476	0.1751	0.1744	0.1781	0.1476
Uniform, c=2, B=144	0.1397	0.1397	0.1343	0.1042	0.1124	0.1124	0.1541	0.1515	0.1607	0.1124
Uniform, c=5, B=48	0.1691	0.1691	0.1679	0.1434	0.1476	0.1476	0.1819	0.1761	0.1850	0.1476
Uniform, c=5, B=144	0.1397	0.1397	0.1343	0.1042	0.1124	0.1124	0.1697	0.1558	0.1765	0.1124
Uniform, c=10, B=48	0.1691	0.1691	0.1679	0.1434	0.1476	0.1476	0.1827	0.1769	0.1861	0.1476
Uniform, c=10, B=144	0.1397	0.1397	0.1343	0.1042	0.1124	0.1124	0.1753	0.1582	0.1838	0.1124
Tilt 0.25, c=5, B=48	0.1958	0.2034	0.1891	0.1822	0.1797	0.1985	0.1970	0.1973	0.1981	0.1888
Tilt 0.25, c=5, B=144	0.1813	0.1929	0.1766	0.1782	0.1570	0.1894	0.1852	0.1863	0.1866	0.1755
Tilt 0.50, c=2, B=48	0.1974	0.2166	0.1983	0.1896	0.1804	0.2081	0.1927	0.1938	0.1919	0.1884
Tilt 0.50, c=2, B=144	0.1848	0.2103	0.1967	0.1885	0.1615	0.2008	0.1780	0.1810	0.1718	0.1626
Tilt 0.50, c=5, B=48	0.1974	0.2166	0.1983	0.1896	0.1804	0.2081	0.1973	0.1985	0.1995	0.1941
Tilt 0.50, c=5, B=144	0.1848	0.2103	0.1967	0.1885	0.1615	0.2008	0.1893	0.1884	0.1899	0.1857
Tilt 0.50, c=10, B=48	0.1974	0.2166	0.1983	0.1896	0.1804	0.2081	0.1999	0.1981	0.2021	0.1949
Tilt 0.50, c=10, B=144	0.1848	0.2103	0.1967	0.1885	0.1615	0.2008	0.1934	0.1897	0.1953	0.1933
Weighted tilt, c=5, B=48	0.2071	0.2252	0.2046	0.2008	0.1929	0.2194	0.2086	0.2083	0.2099	0.2039
Weighted tilt, c=5, B=144	0.1965	0.2200	0.2020	0.1996	0.1773	0.2129	0.1990	0.2008	0.1995	0.1959
Biased validation, c=5, B=48	0.1975	0.2167	0.1960	0.1894	0.1802	0.2061	0.2051	0.2006	0.2064	0.2041
Biased validation, c=5, B=144	0.1848	0.2093	0.1950	0.1880	0.1612	0.1994	0.2081	0.1998	0.2114	0.2226